

ROS2 Application

System configuration: ubuntu22.04

ROS2 Version: humble

1. ROS2 Environment Configuration

a. Update download source

代码块

```
1 sudo apt update
```

b. Enter the ros2 download command

代码块

```
1 wget http://fishros.com/install -O fishros && . fishros
```

2. Device Connected to Virtual Machine

a. View Device

代码块

```
1 ll /dev/ttyUSB*
```

b. Establish Port Mapping

代码块

```
1 sudo gedit /etc/udev/rules.d/99-serial-imu.rules
```

c. Fill in the content of the mapping file

代码块

```
1 KERNEL=="ttyUSB*", ATTRS{idVendor}=="1a86", ATTRS{idProduct}=="7523",
```

```
MODE:="0777", SYMLINK+="imu-serial"
```

d. Save and exit, then run the command to make the rules take effect

代码块

```
1 sudo udevadm trigger
2 sudo service udev reload
3 sudo service udev restart
```

e. Verify

代码块

```
1 ll /dev/imu-serial
```

3. Import the prepared Compressed Packet

- a. Feishu has it in the same level directory [📄 IMU_ROS2.zip](#)
- b. Transferred into the virtual machine via file transfer software
- c. Install the IMU_Library

代码块

```
1 # 下载解压IMU_ROS2压缩文件后, 进入到IMU_Library目录下, 运行setup.py
2 cd IMU_ROS2/IMU_Library
3
4 # 安装库及其依赖
5 pip install -e .
6
7 # 或使用setup.py安装
8 python setup.py install
```

4. Installation of Python-related Libraries

代码块

```
1 sudo pip3 install pyserial
2 sudo pip3 install smbus2
```

5. Build ROS2 project

a. Return to the ~/IMU_ROS2 directory

代码块

```
1 cd IMU_ROS2
2 colcon build --symlink-install
```

a. Write the working directory ~/IMU_ROS2 into the environment variable

代码块

```
1 # 编辑 ~/.bashrc
2 sudo gedit ~/.bashrc
3
4 # 把下面命令写到末尾
5 source ~/IMU_ROS2/install/setup.bash
```

 After successful compilation, execute the following command to check if there are any executable files under the imu_ros2 package

```
ros2 pkg executables imu_ros2
```

6. Start ROS2 node

代码块

```
1 source install/setup.bash
2 ros2 run imu_ros2 imu_publisher
```

7. IMU Data Printing

a. Open a new terminal and view the imu topic

代码块

```
1 ros2 topic list
```

b. Print the data of the /imu/data topic

代码块

```
1  ros2 topic echo /imu/data
```

c. Open a new terminal and view the msg topic

代码块

```
1  ros2 topic echo /imu/mag
```

8. RViz2 Visualization

a. Run the command to open the rviz visualization interface

代码块

```
1  ros2 launch imu_ros2 imu_visualization.launch.py
```

9. Frequently Asked Questions

a. If the startup node fails to open, please try the following commands

代码块

```
1  # 在~/IMU_ROS2目录下运行
2  source install/setup.bash
3
4  # 端口号问题
5  sudo chmod 666 /dev/imu-serial
```